

- Explain what the transformer does to the HTML retrieved by the loader

It walks through the html and extracts every text that is contained within an article, and “cleans” it, which is then returned to the calling function

- Examine the document cleaning code. What kinds of characters are removed when the text is cleaned?

The clean_text function mainly just removes leading and trailing white space, multiple line spacing, and accents/special characters.

- What size chunks is the content split into?

10000 characters

- What amount of overlap is there between chunks?

1000 characters

- How many documents are loaded at a time into the vector database?

300

- Tell me about the MS program

<https://www.pdx.edu/computer-science/master>

- Tell me about the Cybersecurity certificate

<https://www.pdx.edu/computer-science/cybersecurity>

- Tell me about PCEP

<https://www.pdx.edu/computer-science/pcep-internship>

- Tell me about the capstone

<https://www.pdx.edu/computer-science/bachelors-program-computer-science-capstone>

- How many credits does it take to complete the MS program?

45 credits

```
llm>> How many credits does it take to complete the MS program?
It takes 45 credits to complete the MS program. There is also a thesis option.

For more information, please visit these URLs:

* https://www.pdx.edu/computer-science/cybersecurity
* https://www.pdx.edu/computer-science/master
* https://www.pdx.edu/computer-science/honors-track
llm>>
```

- How many credits does it take to complete the MS program?

```
llm>> How many core courses are there in the Cybersecurity certificate?
There are two core courses in the Cybersecurity Graduate Certificate program. These courses total 6 credit hours.

For more information, please visit the following URLs:

* [https://www.pdx.edu/computer-science/graduate](https://www.pdx.edu/computer-science/graduate)
* [https://www.pdx.edu/computer-science/masters-track-courses](https://www.pdx.edu/computer-science/masters-track-courses)
* [https://www.pdx.edu/computer-science/cybersecurity](https://www.pdx.edu/computer-science/cybersecurity)
llm>>
```

- How many hours per week does a student work in PCEP?

```
* [https://www.pdx.edu/computer-science/cybersecurity](https://www.pdx.edu/c
llm>> How many hours per week does a student work in PCEP?
Students in the PCEP program work 20 hours per week.

For more information, please visit these URLs:

* https://www.pdx.edu/computer-science/pcep-internship
* https://www.pdx.edu/computer-science/internships-and-research
llm>>
```

- How long is the capstone?

```
* https://www.pdx.edu/computer-science/internships-and-research
llm>> How long is the capstone?
The capstone project lasts for two quarters, which is six months.

For more information, please visit the following URLs:

* Capstone home page: https://sites.google.com/pdx.edu/bruceirvin/capstone
* Guidelines for Sponsors: https://sites.google.com/pdx.edu/bruceirvin/capstone
* Bachelor's Program in Computer Science Capstone: https://www.pdx.edu/computer-science/bachelors-program-computer-science-capstone
* Undergraduate Courses: https://www.pdx.edu/computer-science/undergraduate-courses
llm>> |
```

- What is the current xkcd?

```
> Finished chain.

Observation: The current XKCD comic is number 3061, titled "Water Balloons". The image URL is https://imgs.xkcd.com/comics/water_balloons.png, the alt text is "Update: The physics department has recruited an astronomer who studies meteor fireballs.", and there is no transcript.
Thought: I am finished executing a plan and have the information the user asked for.
Final Answer: The current XKCD comic is number 3061, titled "Water Balloons". The image URL is https://imgs.xkcd.com/comics/water_balloons.png, the alt text is "Update: The physics department has recruited an astronomer who studies meteor fireballs.", and there is no transcript.

> Finished chain.
{'input': 'What is the current xkcd?', 'output': 'The current XKCD comic is number 3061, titled "Water Balloons". The image URL is https://imgs.xkcd.com/comics/water_balloons.png, the alt text is "Update: The physics department has recruited an astronomer who studies meteor fireballs.", and there is no transcript.'}
llm>> |
```

- What is the image link of the current xkcd?

https://imgs.xkcd.com/comics/water_balloons.png,

- What was xkcd 327 about?

```
> Finished chain.
{'input': 'What was xkcd 327 about?', 'output': 'xkcd 327 was titled "Exploits of a Mom". The alt text is "Help I\'m trapped in a driver\'s license factory." The image URL is https://imgs.xkcd.com/comics/a_mom.png'}
llm>> |
```

- Take a screenshot of the output to include in your lab notebook. How many networks, subnetworks, and VM instances have been created?

1 network, 5 subnetworks, and 5 VM instances

NAME: asia-east1
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: asial-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: e1-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: eu1-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: europe-west1
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: networking101
TYPE: compute.v1.network
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: us-east5
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: us-west-s1
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: us-west-s2
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: w1-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: w2-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

- Visit the web console for VPC network and show the network and the subnetworks that have been created. Validate that it has created the infrastructure in the initial figure. Note the lack of firewall rules that have been created.

[←](#) VPC network details [DELETE VPC NETWORK](#)

networking101

OVERVIEW SUBNETS STATIC INTERNAL IP ADDRESSES FIREWALLS FIREWALL ENDPOINTS ROUTES VPC NETWORK PEER

Subnets [+ ADD SUBNET](#) [MANAGE FLOW LOGS](#)

Filter Enter property name or value

☐	Name ↑	Region	Stack Type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal range
☐	asia-east1	asia-east1	IPv4 (single-stack)	10.40.0.0/16			None
☐	europe-west1	europe-west1	IPv4 (single-stack)	10.30.0.0/16			None
☐	us-east5	us-east5	IPv4 (single-stack)	10.20.0.0/16			None
☐	us-west-s1	us-west1	IPv4 (single-stack)	10.10.0.0/16			None
☐	us-west-s2	us-west1	IPv4 (single-stack)	10.11.0.0/16			None

Reserved proxy-only subnets for load balancing

☐	Name	Region ↑	IP address ranges	Gateway	Role	Purpose
No rows to display						

[EQUIVALENT REST](#)

- Visit the web console for Compute Engine and show all VMs that have been created, their internal IP addresses and the subnetworks they have been instantiated on. Validate that it has created the infrastructure shown in the initial figure.

VM instances

CREATE INSTANCE

IMPORT VM

REFRESH

INSTANCES

OBSERVABILITY

INSTANCE SCHEDULES

Your project's VMs use global DNS names by default. To reduce the risk of cross-regional outages, we recommend you use zonal DNS instead. [Learn more](#)

VM instances

Filter

Enter property name or value

☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Network	Connect
☐		asia1-vm	asia-east1-b			10.40.0.2 (nic0)	35.194.242.159 (nic0)	networking101	SSH
☐		course-vm	us-west1-b			10.138.0.2 (nic0)		default	SSH
☐		e1-vm	us-east5-a			10.20.0.2 (nic0)	34.162.113.167 (nic0)	networking101	SSH
☐		eu1-vm	europa-west1-d			10.30.0.2 (nic0)	104.199.74.211 (nic0)	networking101	SSH
☐		guestbook	us-west1-b			10.138.0.18 (nic0)		default	SSH
☐		guestbook2	us-west1-a			10.138.0.20 (nic0)		default	SSH
☐		vm-europe-west1-d	europa-west1-d			10.132.0.2 (nic0)		default	SSH
☐		w1-vm	us-west1-a			10.10.0.2 (nic0)	35.185.251.162 (nic0)	networking101	SSH
☐		w2-vm	us-west1-a			10.11.0.100 (nic0)	34.82.105.150 (nic0)	networking101	SSH

Related actions

Explore Backup and DR NEW

Back up your VMs and set up disaster recovery

View billing report

View and manage your Compute Engine billing

Monitor VMs

View outlier VMs across metrics like CPU and network

Explore VM logs

View, search, analyze, and download VM instance logs

Set up firewall rules

Control traffic to and from a VM instance

- Click on the **ssh** button for one of the VMs and attempt to connect. Did it succeed?

No! It is stuck on “Establishing connection to SSH server...”

- Take a screenshot that indicates the new rules have been deployed

networking101

[<](#)
[OVERVIEW](#)
[SUBNETS](#)
[STATIC INTERNAL IP ADDRESSES](#)
[FIREWALLS](#)
[FIREWALL ENDPOINT](#)

[ADD FIREWALL RULE](#)
[DELETE](#)

Filter

Enter property name or value

☐	Name	Enforcement order ↑	Type	Deployment scope
	▼ vpc-firewall-rules	1	VPC firewall rules	Global
☐	networking-firewall-allow-ssh		Ingress firewall rule	Global
☐	networking-firewall-allow-internal		Ingress firewall rule	Global
☐	networking-firewall-allow-icmp		Ingress firewall rule	Global

[EQUIVALENT REST](#)

- Given this, fill in the table with the measured latencies between the 6 pairs and include it in your lab notebook. Use the shortest latency measured for each pair.

Location pair	Ideal latency	Measured latency
us-west1 us-east5	~45 ms	53.9ms
us-west1 europe-west1	~93 ms	102ms
us-west1 asia-east1	~114 ms	116 ms
us-east5 europe-west1	~76 ms	91.4ms

us-east5 asia-east1	~141 ms	
		174 ms

europa-west1	~110 ms	
asia-east1		264 ms

- Are the instances in the same availability zone or in different ones?

Different!

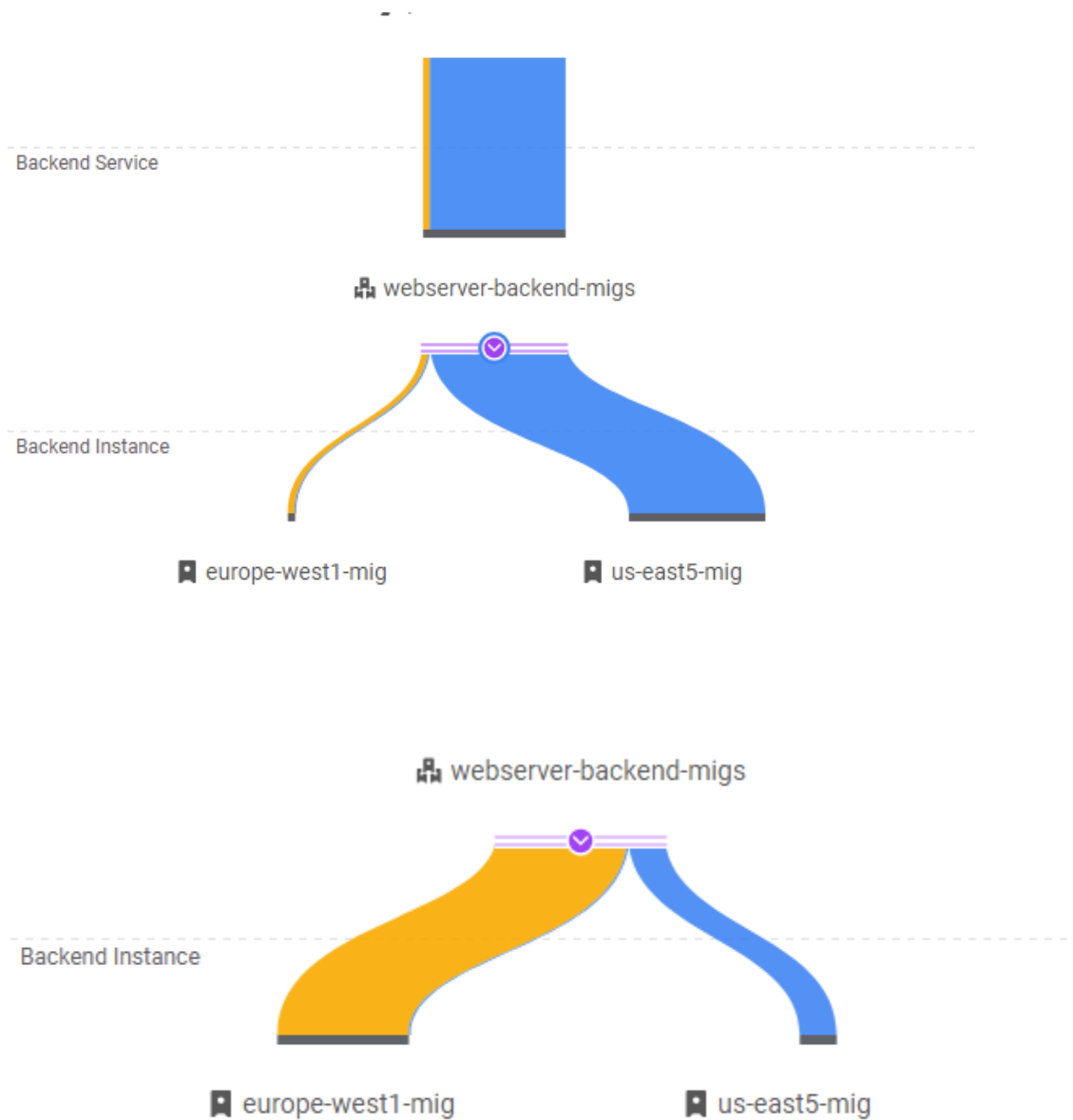
- List all availability zones that your servers show up in for your lab notebook.

Europe-west1-c, europe-west1-b, europe-west1-d, us-east-5b

- Which availability zone does the server handling your request reside in?

Us-east-5b

- Take a screenshot of the initial traffic distribution



- Take a screenshot of the UI as additional instances are brought up and show that the traffic distribution shifts

